

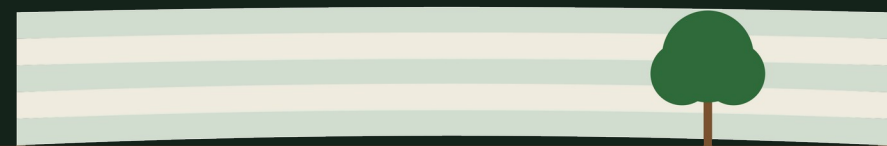
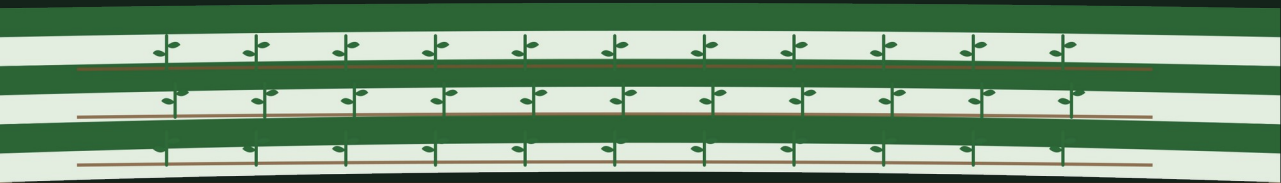
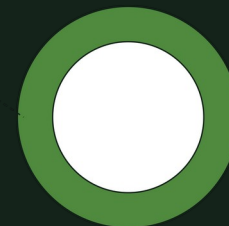


DEALER TRAINING · SEPTEMBER 2026

Cala-Mag V

Calcium, magnesium and boron in 1 acidic liquid, with nitrogen: N 10, Ca 6, Mg 4, B 1 grams per 100 ml.

For VAN dealers and the field team, written so a dealer can repeat it to a grower. Lahore.



What a dealer says in 30 seconds.

Cala-Mag V is calcium, magnesium and boron in 1 bottle, with nitrogen: N 10, Ca 6, Mg 4 and B 1, grams in every 100 ml, acidic so the calcium stays in solution. Pakistani soil is full of calcium as lime, and the plant still runs short, because calcium in lime and calcium the root can drink are 2 different things, and a heavy potash dose blocks it at the root. Calcium does not move inside the plant once it is placed, so the fruit, the tuber and the growing tip fail first: blossom end rot on tomato, tip burn, hollow heart in potato. It has to arrive while the fruit is being built, and it has to be repeated. The programmes put it on tomato and chili as a spray at early and grand growth, on potato 4 bottles through the water at germination and again at grand growth, on strawberry, mango and guava through the water from flowering through fruiting.

The 4 things it must contain

1

The bottle: 1 L, N 10, Ca 6, Mg 4, B 1, PS 5469-2020. 60 g of calcium and 40 g of magnesium in every litre.

2

The paradox: the soil is full of lime and the plant still runs short. Heavy potash makes it worse.

3

The window: calcium does not move inside the plant. It must arrive while the fruit is built, repeated, from flowering through fill.

4

The routes: spray on tomato and chili; through the water on potato, strawberry, mango and guava. The row says which.

Say it in the grower's own words. Every figure in these 30 seconds is on the bag, in the published crop programme, or in the named evidence on the slides that follow.

THE BAG

What the bottle says, and what it is registered as.

1 L

the bottle. 100 g of N, 60 g of Ca, 40 g of Mg and 10 g of B in every litre, from the declared analysis.

Ca 6 • Mg 4

w/v: 6 g of calcium and 4 g of magnesium in every 100 ml, in a ratio set so neither blocks the other, as VAN's product data says.

N 10

nitrogen, 100 g per litre, to keep the plant working while the fruit is built.

B 1

boron, 10 g per litre, as VAN publishes the analysis. It travels with the calcium to the growing point.

pH 2 to 4

acidic, transparent, 100% soluble, as VAN's product data gives it. Calcium drops out of solution as it approaches neutral; the acid is what keeps it liquid.

2 routes

fertigation and foliar, as VAN names them. The programme row says which for your crop; guava prints the spray rate, 500 ml per 100 L of water.



Registered as	Pakistan Standard	Manufacturing licence
Cala-Mag V	PS 5469-2020	CM/L-4299/2026

No expiry date is printed in this deck. The licence number is what a dealer quotes.

A GROWER'S QUESTION

“My tomatoes came out with a black sunken patch at the bottom. The sprayer said it was a disease. 3 sprays later it was still there. What is it?”

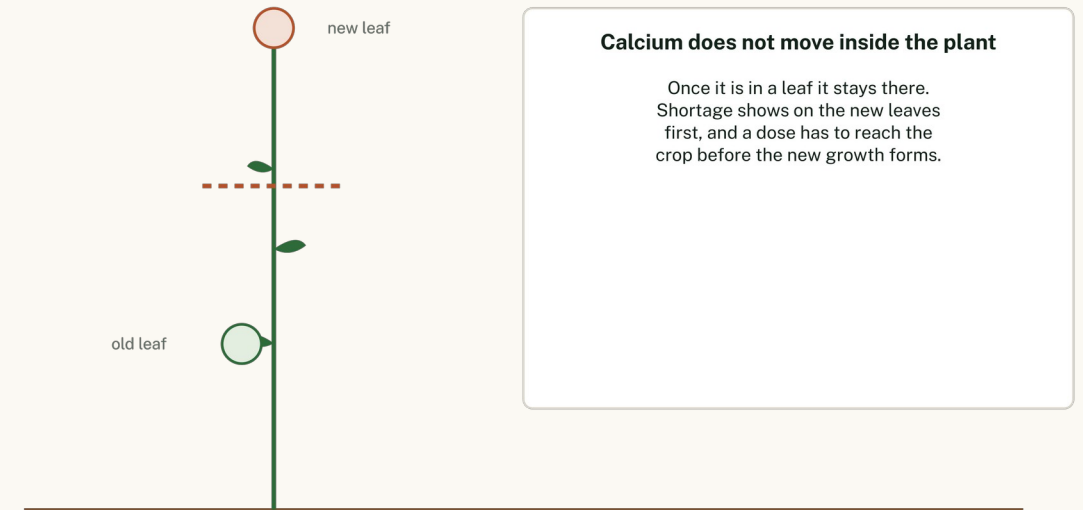
It is calcium that did not arrive while the fruit was forming, and no fungicide will put it there. As VAN tells its own story, the product began the same way: a tomato crop in Sindh with patches on the fruit being treated for a virus, which VAN's team read as a calcium shortage on soil that was full of calcium. The answer is 3 ideas: why a soil full of lime still starves the fruit, why the window does not reopen, and what is in the bottle that lets calcium, magnesium and boron travel together.

The 3 answers on the next 3 slides, 1 idea each.

The soil is full of lime. The plant still runs short.

Calcium is the cement of the cell wall and the tip of every growing point. Punjab soil carries on average 6.53% calcium carbonate, lime, and the root cannot drink lime; it drinks the free calcium in the soil water, and a heavy potash or magnesium dose crowds that out at the root. Inside the plant, calcium does not move once it is placed. So the oldest leaves look fine and the youngest tissue fails: blossom end rot on tomato, tip burn on leafy crops, hollow heart in potato. Magnesium is the opposite: it moves, so the old leaves go yellow between the veins first.

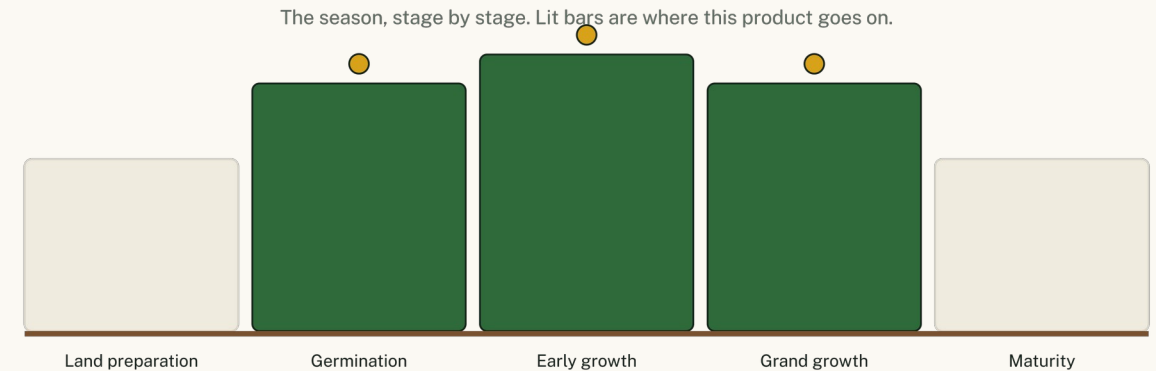
Say it like this: “Your soil is full of calcium. Your fruit got none. They are 2 different things.”



Calcium drawn as immobile inside the plant: what the fruit did not receive while it was forming, it never receives. General plant nutrition, stated as such.

It must arrive while the fruit is built. The window does not reopen.

Because calcium does not redistribute inside the plant, the fruit and the tuber take it only while they are forming, from flowering through fill. Calcium arriving after the fruit has formed cannot move into it. So the programmes apply it inside the window and repeat it: tomato and chili at early growth and grand growth; potato at germination and again at grand growth; strawberry at early growth and grand growth; mango at grand growth; guava in September, at flowering in October with a repeat 15 days later, and at fruiting in November with a repeat 15 days later. Several modest applications beat 1 large one, and that is a consequence of the chemistry, not a sales preference.

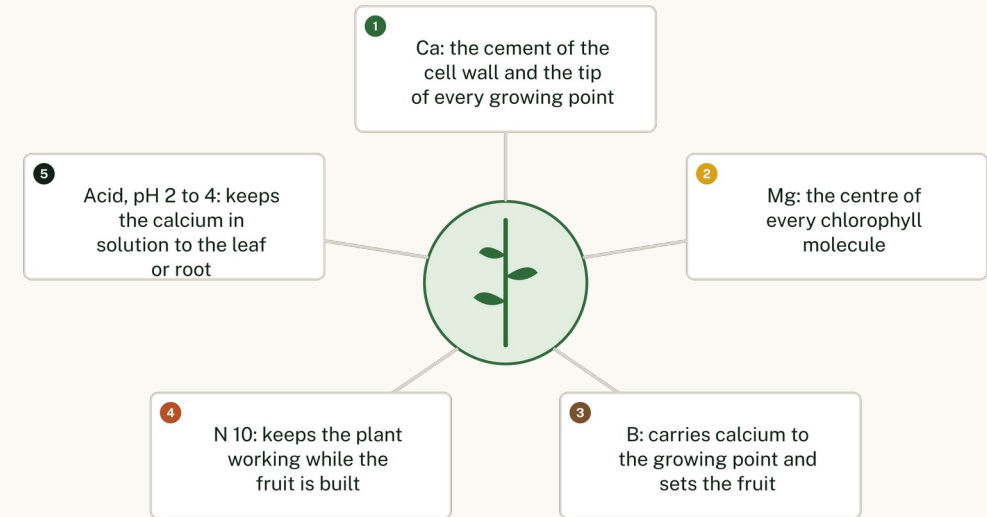


The stages the programmes name. Potato takes it at germination; the rest from early growth. The rows are on the programme slides.

Say it like this: “Give it while the fruit is being made, more than once. After the fruit is formed it is too late.”

Calcium, magnesium and boron in 1 bottle, kept in solution.

Calcium and magnesium compete for the same uptake sites, so a careless mix of the 2 lets each block the other; VAN's product data says the ratio in this bottle is set so that neither does. Boron travels with calcium: it is needed for calcium to reach the growing point and to build the wall, and it sets the flower and the fruit. The carrier is acidic, pH 2 to 4, because calcium precipitates as a solution approaches neutral, which is exactly what the lime in the soil does to it. Amino acids are in the formulation to help the 3 move through the leaf surface rather than sit on it. The nitrogen keeps the plant working through the window.



The 5 parts of the litre, from VAN's product data. The amino acids carry no label figure and are named, not numbered.

Say it like this: "3 nutrients that fight each other in the soil, carried together in 1 bottle that keeps them liquid."

THE EVIDENCE

What is verified for Cala-Mag V, and what is not.

No trial of Cala-Mag V is on VAN's record. What a dealer may state as fact is the bottle, the licence, the programme rows, the soil record and the chemistry, each as what it is.

Item	What it says	How a dealer may use it
The bottle	Cala-Mag V, N 10%, Ca 6%, Mg 4%, B 1% w/v; transparent liquid, pH 2 to 4, 100% soluble, amino acids in the formulation; 1 L.	As fact. Read the analysis off the bottle in hand.
The licence	Pakistan Standard PS 5469-2020 for liquid nutrient supplements; manufacturing licence CM/L-4299/2026.	As fact. The number, never an expiry date.
The programme rows	11 rows on 6 programmes. Spray: chili at grand growth, tomato at early and grand growth. Fertigation: potato 4 packs at germination and 4 at grand growth; strawberry 4 at early growth and 1 at grand growth; mango 1 at grand growth; guava 1 in September (or spray at 500 ml per 100 L), 1 at flowering and 1 at fruiting, each repeated 15 days later.	As the published rate, as printed.
The soil record (Punjab, 770,160 samples)	Mean calcium carbonate 6.53%; 69.1% of samples above pH 8; 51.7% low in boron; 87.8% below the potash threshold.	As fact about the soil: full of lime, short of boron. Never as product performance.
The chemistry	Calcium is immobile in the plant; magnesium is mobile. Calcium and magnesium compete at the root; heavy potash suppresses both. Calcium precipitates as a solution approaches neutral.	As general plant nutrition, said as such.
Not on file	A Cala-Mag V trial; a yield, grade, firmness or shelf-life figure; a blossom-end-rot cure rate; an amino acid figure; a compatibility chart; a Pakistani figure for how much of the country's soil is calcareous.	Not said until named.

Sources: the bottle; the Pakistan Standard and licence register; the crop programmes on van.com.pk; the VAN farmer price sheet, Aug 2026; the Punjab soil record (770,160 samples); VAN's 17-nutrients page.

The bottle, in grams, in the grower’s own back-of-envelope.

Price from the VAN farmer price sheet, Aug 2026, delivered farmer price. Nutrients from the declared analysis, w/v: 6% is 60 g in a litre.

Cala-Mag V, N 10 · Ca 6 · Mg 4 · B 1	Volume	N	Ca	Mg	B	Rupees
1 bottle	1 L	100 g	60 g	40 g	10 g	990
Tomato, 2 bottles, sprayed	2 L	200 g	120 g	80 g	20 g	1,980
Potato, 8 bottles, through the water	8 L	800 g	480 g	320 g	80 g	7,920

On tomato the grower spends 1,980 rupees an acre to put 120 g of calcium, 80 g of magnesium and 20 g of boron on the leaf across the 2 stages the fruit is built. On potato it is 8 bottles through the water, 7,920 rupees, 480 g of calcium to the tuber zone. Grams, at the window. What the crop does with it is not a number VAN prints.

What the programme puts on, per acre, per season.

Bottles per acre from the published programmes; rupees at the VAN farmer price sheet, Aug 2026. Route as printed.

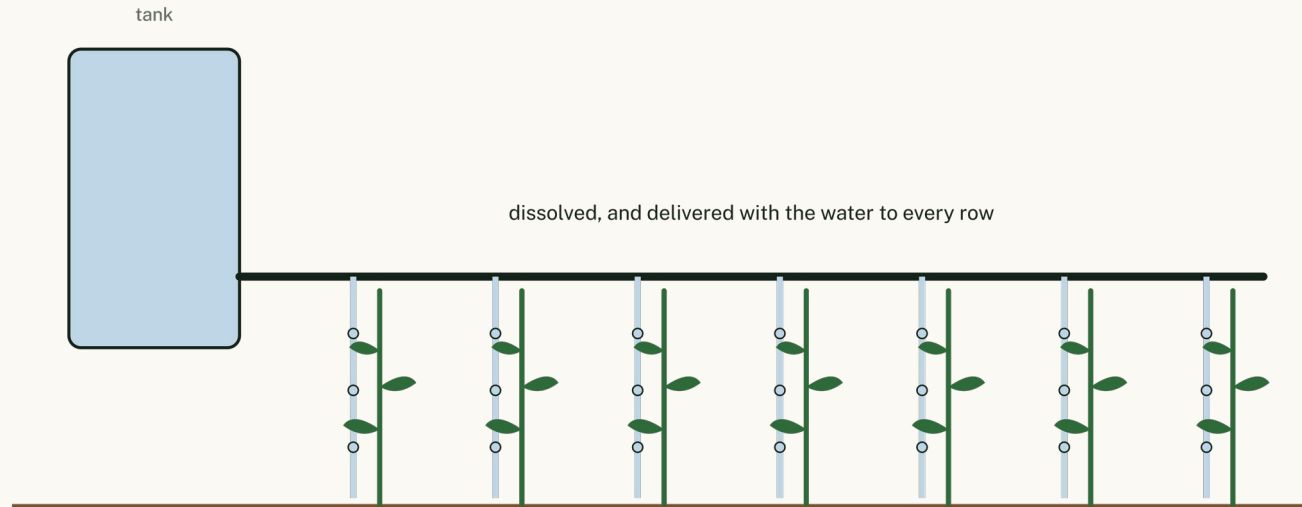
Crop	Stage and method, as printed	Bottles	Rupees	Ca on the acre
Chili	Grand growth, spray	1	990	60 g
Tomato	Early growth, spray; grand growth, spray	2	1,980	120 g
Potato	Germination, 4 packs, fertigation; grand growth, 4 packs, fertigation	8	7,920	480 g
Strawberry	Early growth, 4 packs, fertigation; grand growth, 1 pack, fertigation	5	4,950	300 g
Mango	Grand growth, fertigation	1	990	60 g
Guava	September, fertigation or spray at 500 ml per 100 L; October flowering and November fruiting, fertigation, each repeated 15 days later	3 to 5	2,970 to 4,950	180 to 300 g

Potato is the heaviest programme, because the tuber is built underground across weeks and calcium reaches it only through the water. Guava is the 1 row that prints a spray rate: 500 ml per 100 L of water.

Rows from the crop programmes on van.com.pk, as printed; 1 L at PKR 990. Guava: 3 rows with "repeat 15 days after flowering" and "after fruiting" on 2 of them, read as 3 to 5 bottles. Ca per bottle 60 g.

HOW IT GOES ON

Through the water, or on the leaf. Inside the window, repeated. Do not neutralise it.



Fertigation: dissolve, then release with the water flowing

Potato, strawberry, mango and guava rows. Into the drum at the head of the watercourse or the drip tank, stirred in, released across the whole irrigation and not in 1 slug, into the wet root zone. It is fully soluble. Do not tank-mix it towards neutral, and do not put a phosphate or sulfate fertilizer in the same tank: calcium drops out with either. Jar-test anything else.

Spray: water first, product second, morning or evening

Chili and tomato rows. Fill the tank with clean water, add the bottle, keep it agitated. The guava row prints the rate a plan names: 500 ml per 100 L of water. A fine, even fan, to cover the leaf and the young fruit without run-off. Never midday.

Inside the window, more than once

From flowering through fruit fill on fruiting crops; potato at germination and again at grand growth; guava with the 15-day repeats. Calcium arriving after the fruit has formed cannot move into it, so several applications across the window beat 1 large one. The programme's rate is the rate: boron rides in every bottle, and the gap between enough and too much boron is narrow.

“My soil is full of calcium. Why would I buy calcium?”

The dealer says, out loud

Because the calcium in your soil is lime, and the root does not drink lime. It drinks the free calcium in the soil water, and when you push potash the potash crowds it out at the root. Inside the plant calcium cannot move, so the fruit and the tuber, which are built last, are the first to go short. This bottle puts calcium, with magnesium and boron, where the fruit is forming, while it is forming. 1 bottle, 990 rupees, sprayed on tomato at early growth and again at grand growth.

The fact behind it

- Calcium in lime and calcium the root can drink are 2 different things (VAN's 17-nutrients page).
- Mean calcium carbonate 6.53% across 770,160 Punjab samples (the Punjab soil record).
- Cala-Mag V 1 L 990 rupees (VAN farmer price sheet, Aug 2026); tomato 2 bottles, early and grand growth, spray (the crop programmes on van.com.pk).

Never say: “It gives you X more maunds”, “firmer fruit by X%”, or a rupees-per-acre return. No trial exists.

“Can I put it in the tank with my potash or my phosphate?”

The dealer says, out loud

Not with a phosphate or a sulfate: calcium drops out of solution with either, and you lose the calcium and may block the line. That is basic chemistry, not a fault of the bottle. Run it on its own pass, or jar-test first if you must combine. And keep the acid: the bottle is pH 2 to 4 because calcium precipitates as it approaches neutral, so do not add anything that neutralises it.

The fact behind it

- VAN’s guidance: the acidity is doing work; do not tank-mix towards neutral.
- Calcium phosphate and calcium sulfate are insoluble; they settle out in a tank (textbook chemistry).
- No compatibility chart for Cala-Mag V is published; jar-test anything else.

Never say: “Mix it with anything” or a compatibility list VAN has not published.

“Which stage, and how many bottles?”

The dealer says, out loud

What the plan prints. Tomato: 1 bottle sprayed at early growth and 1 at grand growth. Chili: 1 at grand growth. Potato: 4 bottles through the water at germination and 4 again at grand growth. Strawberry: 4 at early growth and 1 at grand growth, through the water. Mango: 1 at grand growth. Guava: 1 in September, 1 at flowering in October and 1 at fruiting in November, each repeated 15 days later. Not after the fruit has formed: by then the calcium cannot get in.

The fact behind it

- Tomato 2 × 1 pack, spray; chili 1 pack, spray; mango 1 pack, fertigation (the crop programmes on van.com.pk).
- Potato 4 + 4 packs, fertigation; strawberry 4 + 1 packs, fertigation; guava 3 rows with 15-day repeats (the crop programmes on van.com.pk).
- Timing beats rate; repeat, do not concentrate (VAN guidance).

Never say: “1 big dose when you see the rot.” Or a rate for cotton or orchards, which have no row.

“Spray or through the water? Which is better?”

The dealer says, out loud

The row for your crop says which, and the plan is built that way for a reason. Tomato and chili are sprayed, onto the leaf and the young fruit, where the calcium is needed and cannot travel to. Potato, strawberry, mango and guava take it through the water, because the tuber and the root zone are where it has to land. Guava gives you both: fertigation, or a spray at 500 ml per 100 L of water.

The fact behind it

- VAN names fertigation and foliar as the routes for Cala-Mag V.
- Spray on chili and tomato; fertigation on potato, strawberry, mango and guava; both on the guava September row (the crop programmes on van.com.pk).
- Calcium is immobile in the plant: it has to land where the fruit is forming (VAN's 17-nutrients page).

Never say: A spray rate for a fertigation row, or the reverse, from memory.

“The rot is already on the fruit. Will this cure it?”

The dealer says, out loud

No bottle puts calcium into a fruit that has already formed, and VAN does not say this one does. The fruit with the patch is lost. What the spray does is reach the flowers and young fruit forming now, at the stage the plan names, so the next set is built with calcium in the wall. Put it on now and again at the next stage. Not a cure; a supply for the fruit still to come.

The fact behind it

- Calcium does not move inside the plant once placed; the youngest tissue fails first (VAN's 17-nutrients page).
- Calcium arriving after the fruit has formed cannot move into it (VAN guidance; textbook plant nutrition).
- No cure rate for blossom end rot is on VAN's record.

Never say: “It cures blossom end rot” or “it stops the rot in a week”. No cure rate exists.

“There is boron in it. Will it burn my crop?”

The dealer says, out loud

Not at the programme's rate. Boron is needed in grams an acre and the gap between enough and too much is narrow, which is exactly why it is carried inside this bottle at a fixed share instead of dosed loose. 1 bottle carries 10 g of boron; the tomato programme puts 20 g on the acre across the season. Use the plan's bottle count and no more. Extra bottles are extra boron the crop did not ask for.

The fact behind it

- The narrow window between enough and too much boron (VAN's 17-nutrients page).
- B 1% = 10 g per litre; tomato 2 bottles = 20 g; potato 8 bottles = 80 g (the declared analysis and the crop programmes on van.com.pk).
- 51.7% of Punjab samples read low in boron (the Punjab soil record).

Never say: “Boron cannot harm a crop” or “double the bottles on a boron-short field”.

“Is it registered? What is the licence?”

The dealer says, out loud

Pakistan Standard PS 5469-2020 for liquid nutrient supplements, manufacturing licence CM/L-4299/2026, VAN as the registered maker. Every batch is tested at VAN Lab, PNAC LAB 336, before it ships. Read the batch number off the bottle and VAN sends you that batch's certificate, free.

The fact behind it

- Cala-Mag V: PS 5469-2020, CM/L-4299/2026 (the licence register on van.com.pk).
- Every batch is tested at VAN Lab, PNAC LAB 336, before it ships.
- The certificate for any batch is free on request against the batch number.

Never say: An expiry date, or an amino acid figure as if it were on the certificate.

Cala-Mag V in the published programmes, per acre.

Crop	Stage	Rate per acre	Method	Pack
Tomato	Early Growth	1 pack	Spray	1 L
Tomato	Grand Growth	1 pack	Spray	1 L
Chili	Grand Growth	1 pack	Spray	1 L
Potato	Germination	4 Pack	Fertigation	1 L
Potato	Grand Growth	4 Pack	Fertigation	1 L
Strawberry	Early Growth	4 pack	Fertigation	1 L

Rows as published in van.com.pk's crop programmes, all 6 programmes that name Cala-Mag V; spray on tomato and chili, fertigation on the other 4, both on the guava September row. A rate is per acre; the crop page scales it to the field.

Cala-Mag V in the published programmes, continued.

Crop	Stage	Rate per acre	Method	Pack
Strawberry	Grand Growth	1 pack	Fertigation	1 L
Mango	Grand Growth	1 pack	Fertigation	1 L
Guava	September	1 pack	Fertigation, or spray at 500 ml per 100 L water	1 L
Guava	October, flowering starts	1 pack	Fertigation. Repeat 15 days after flowering	1 L
Guava	November, fruiting	1 pack	Fertigation. Repeat 15 days after fruiting	1 L

Rows as published in van.com.pk's crop programmes, all 6 programmes that name Cala-Mag V; spray on tomato and chili, fertigation on the other 4, both on the guava September row. A rate is per acre; the crop page scales it to the field.

WHAT VAN WILL NOT CLAIM FOR IT

A dealer who says any of these is speaking for himself, not for VAN.

1 A yield, grade, firmness or shelf-life figure. No Cala-Mag V trial exists; "grade, not tonnage" stays a description of what calcium does, never a number.

2 That it cures blossom end rot, tip burn or hollow heart, or any disease. Calcium cannot enter a fruit already formed.

3 An amino acid figure, or any figure the bottle does not declare, as if it were on the certificate.

4 A rate for cotton or orchards. They have no programme row.

5 That it fixes a soil that is short of calcium, or changes the soil's calcium or pH. It supplies the plant through the window; a soil test measures the field.

6 A compatibility promise. Not with phosphate or sulfate; jar-test the rest; no chart is on file.

Every batch is tested. The bag carries the proof.

1 · The batch number

It is printed on the bag. Read it to the grower and show him where it is. Every bag of every batch carries one.

2 · The certificate of analysis

For any batch, free on request. Send the batch number off the bag to kisan@van.com.pk or the helpline, and the certificate for that batch comes back. No sample to post.

3 · The laboratory

VAN Lab, PNAC LAB 336, accredited to ISO/IEC 17025:2017. Every batch is released through it before it ships. The certificate carries the laboratory's number.

What a certificate for Cala-Mag V shows

The declared analysis tested for the batch: nitrogen, calcium and magnesium against N 10, Ca 6, Mg 4 w/v, with boron where the certificate carries it, the batch number, the Pakistan Standard PS 5469-2020 and licence CM/L-4299/2026 on the bottle. The amino acids carry no label figure and are not a certificate line. No certificate for a Cala-Mag V batch was on file for this deck, so no batch reading is printed here; the grower asks for his own batch's certificate.

The lines a dealer says in Urdu.

The first line is the promise word from the VAN brand master. The other 3 are the deck's key sentences, translated in a first pass and not yet reviewed by VAN. Say them, and send corrections to partner@van.com.pk.

Calcium while the fruit is built. No promise word: calcium has none yet.

Urdu first pass, unreviewed

کیلشیم پھل بنتے وقت چاہیے، بعد میں نہیں

The soil is full of lime, and still the plant gets none.

Urdu first pass, unreviewed

مٹی چونے سے بھری ہے، پھر بھی پودے کو کیلشیم نہیں ملتا

From flowering to fruit fill, a little, more than once.

Urdu first pass, unreviewed

پھول سے پھل بھرنے تک، تھوڑا تھوڑا، بار بار

Heavy potash blocks calcium and magnesium at the root.

Urdu first pass, unreviewed

زیادہ پوٹاش جڑ پر کیلشیم اور میگنیشیم کو روک دیتی ہے

Write to the address for what you need.

If you grow

kisan@van.com.pk

A crop plan, product rates, a certificate for your batch, a dealer near you.

If you sell

partner@van.com.pk

Dealers, distributors and the trade. Stock, training, corrections to this deck.

Everything else

info@van.com.pk

Supply, the laboratory, a plant visit.

Helpline 0300 5003041 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd, Lahore. 1 helpline, 3 addresses. No group broadcasts: a grower who writes gets 1 reply.

Cala-Mag V: calcium, magnesium and boron in 1 bottle, from flowering through fruit fill, repeated. After the fruit is formed it is too late.

